Vector addition

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Add displacement vectors and use a model to explain where an object ends up.

SUCCESS CRITERIA - I can...

1. Connect tip-to-tail diagrams to signed component sums.
2. Give magnitude, units and an angle with a named reference direction.
3. Check the result against the diagram, limits and physical situation.

Core: teacher selects from 1-10. **Stretch:** 11-12 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “vector” and “component” to explain your model or result. Sentence starters are optional.

vector: a quantity with size and direction

component: the part of a vector along one chosen axis

resultant: one vector equal to the sum of other vectors

01 Retrieve and distinguish

A cart moves from 0 m to +6 m in 3 s, then to +2 m in 2 s. **Find** average speed and average velocity for the full trip. **Sketch** the route and explain why the answers differ.

Word help: velocity = rate of position change, with direction

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

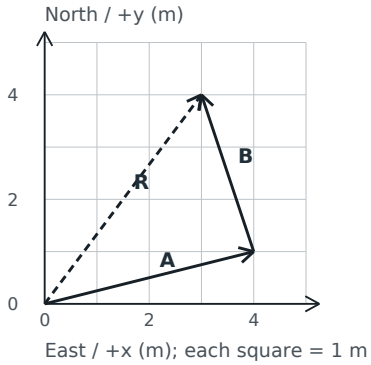
02 Explain a worked model

Reference: $R = \sqrt{(R_x^2 + R_y^2)}$.

Worked reasoning: East=+x; north=+y. R_x=4-1=3 m; R_y=1+3=4 m; R=√(3²+4²)=5 m.
θ=tan⁻¹(4/3)=53.1° north of east.

Your turn: Find B’s westward component. **Explain** why R begins at the original start.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .



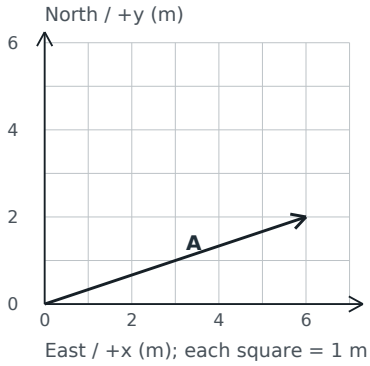
Marzano check - circle: ① ② ③

03 Complete, then check

A=(6, 2) m; B=(-4, 3) m. A is drawn. Add B and R tip to tail; find R’s components, magnitude and direction. **Check** the quadrant.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: ① ② ③

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Continue down each column.

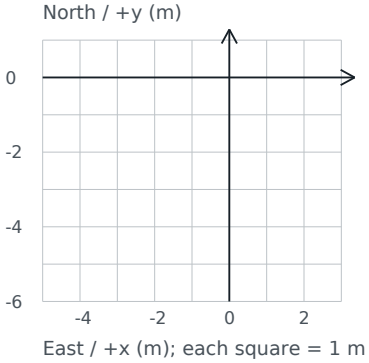
Sentence starters are optional. You may plan in any language.

04 A different quadrant

A = (2, -5) m and B = (-6, 2) m.
Predict the resultant quadrant.
Draw tip to tail; calculate the components, magnitude and direction.
State how you know your angle is in the correct quadrant.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I predict … . My model supports this because … .



Marzano check - circle: 1 2 3

05 Choose the correct component signs

A is 10 m at 30° north of east: A_x = 10 cos 30° = 8.66 m and A_y = 10 sin 30° = 5.00 m.
B is 6 m at 60° north of west.
Sketch both arrows, resolve B into signed components, and find the resultant magnitude and direction.

Word help: resolve = separate a vector into components

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

06 The angle is measured from north

Walk 12 m at 25° east of north, then 7 m west.
Sketch first and predict the resultant quadrant.
Calculate components, magnitude and direction.
Explain which component uses cosine and why.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I predict … . My model supports this because … .

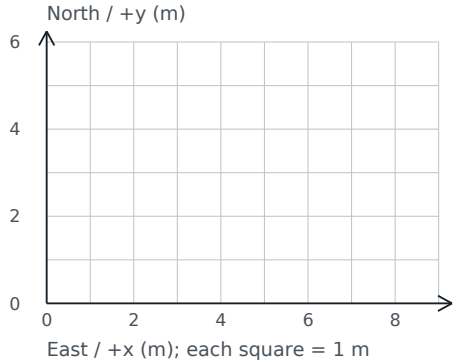
Marzano check - circle: 1 2 3

07 Plan the direct return

A survey team walks 8 m east, 5 m north, then 2 m west. A clear field allows a straight return to the start.
Find the route distance before returning and the direct return displacement, including magnitude and direction.
Draw both the outward route and the return.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3

08 Diagnose the reasoning

A student adds 10 m east and 10 m at 60° north of east using $\sqrt{(10^2 + 10^2)} = 14.1$ m.
Identify the violated condition.
Draw a correct model and use it to repair the solution, including direction.

Sentence starter: The step I would change is … . The evidence is … .

Marzano check - circle: 1 2 3

Vector addition

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Continue down each column.

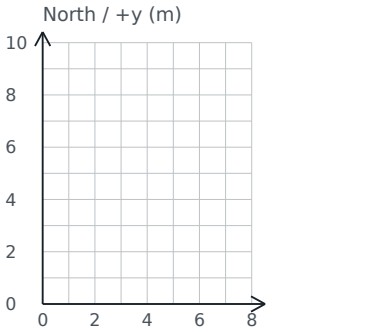
Sentence starters are optional. You may plan in any language.

09 Represent one trip in several ways

A robot moves steadily from (0, 0) m to (6, 0) m in 3 s, then steadily to (6, 8) m in another 4 s.
(a) Draw the path and resultant; find distance and displacement.
(b) Find average speed and the average-velocity vector for the full trip.
(c) Sketch x-versus-time and y-versus-time graphs.
Explain how each graph records the same motion as your path diagram.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: Under the assumption …, the evidence … supports … because … .



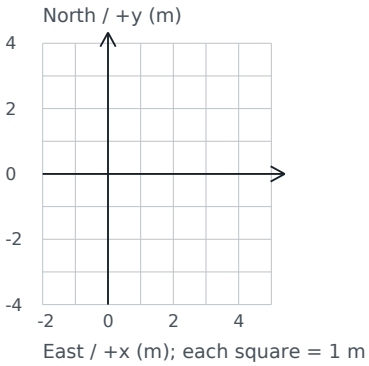
Marzano check - circle: 1 2 3

10 Exit check - solve without the worked model

A = (4, -3) m and B = (-1, 5) m.
Find R, its magnitude and direction.
Choose C so A + B + C = 0.
Draw a closed tip-to-tail route and give one physical or mathematical check of your answer.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.



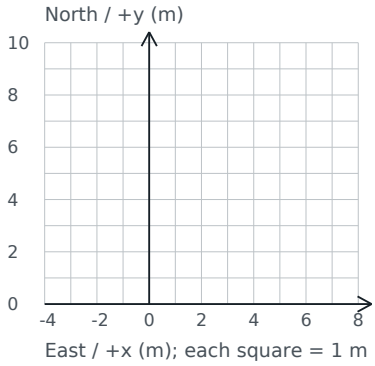
Marzano check - circle: 1 2 3

11 STRETCH: Missing direction

The first displacement is 6 m east. The second has magnitude 10 m. Finish directly north of the start.
Find the second vector's components and direction, and the resultant.
Draw the model.
Explain whether more than one solution meets the north requirement.

Word help: magnitude = size of a quantity without its direction

Sentence starter: Under the assumption …, the evidence … supports … because … .



Marzano check - circle: 1 2 3

12 STRETCH: Decide what is possible

Two displacements have magnitudes 5 m and 3 m.
Test whether resultants of 1 m, 6 m and 9 m are possible using limits.
Draw the 6 m case to scale.
Estimate the angle between the original arrows with tails together.
Explain why the tip-to-tail triangle uses a different interior angle. Optional: derive an angle-resultant relationship to check your drawing.

Word help: derive = show the reasoning or algebra that produces a relationship

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3